Akib Mohammed Khan *Computer Science Engineer, Ph.D. Student*

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RESEARCH FOCUS

Adversarial robustness and reliable evaluation of vision foundation models. I build attacks and defenses for models that are frozen or adapted only at test time, such as DINOv2 anomaly detectors and CLIP under test-time prompt tuning, with an emphasis on defense-aware evaluation and calibrated uncertainty. The same question motivates my optics work with a star phase mask being a physically measurable perturbation of the input that degrades a downstream vision model.

EDUCATION

PhD in Imaging Science, *Rochester Institute of Technology (RIT)* 2023 – present | Rochester, NY, USA

- CGPA: 3.97 out of 4.00
- Advisor: Dr. Bartosz Krawczyk, MLVision Lab
- Research Interest: adversarial robustness, test-time adaptation, vision foundation models, uncertainty estimation, computational imaging, few-shot and continual learning.

B.Sc. in Computer Science and Engineering, *Islamic University of Technology (IUT)* 2018 – 2022 | Gazipur, Bangladesh

- CGPA: 3.95 out of 4.00
- Ranked 3rd out of 69

RESEARCH EXPERIENCE

Graduate Research Assistant, *MLVision Lab, Center for Imaging Science, RIT* 08/2023 – Present | Rochester, NY, USA

- **Adversarial robustness of vision foundation models.** Built a white-box attack framework for non-parametric k-NN memory-bank anomaly detectors via a learned differentiable distance surrogate, and a two-level defense (input-level randomized patch shifting with median aggregation, feature-level residual purification) that recovers +27% image AUROC and +51% pixel PRO over the undefended attacked baseline on MVTec-AD / VisA / MPDD at 4 shots, within 3% of clean accuracy and stable under adaptive attack. Accepted, BMVC 2026.
- **Sampling-free Bayesian uncertainty for vision transformers (Trust-Vista).** Derived the closed-form moment propagation rules that transport the variational posterior mean and variance through multi-head self-attention, layer normalization, pooling, and residual paths, giving calibrated uncertainty in a single forward pass, and ran the full experimental analysis across six benchmarks up to ImageNet-1k: robustness under Gaussian noise, FGSM, PGD, and C&W∞ attack; and calibration by ECE, NLL, and Brier score against MC-dropout, SNGP, and deep ensembles. Under review, IEEE TIP. Supported by NSF CAREER Award No. 2542166.
- **Physics-based deep learning for sensor protection.** Measured the true point spread function of a star phase mask on an optical bench (HDR three-exposure merge, 16-frame averaging, dark and glare subtraction) and generated multi-domain paired training data by tile-wise shift-variant convolution of DIV2K across focus settings; trained and tuned Restormer / NAFNet restoration models on 8×A100 clusters. In collaboration with Naval Research Laboratory (NRL). Supported by ONR Award N00014-23-1-2513.
- Teaching Assistant for Imaging Science Fundamentals and Probability and Statistics in Imaging.

PROFESSIONAL EXPERIENCE

AI/ML Developer (Remote), *MRIMath LLC* 05/2025 – 08/2025 | Vestavia, AL, USA

- Built and validated deep learning models for **medical image segmentation** across multiple imaging modalities; established reproducible experimental protocols and quantitative evaluation metrics for clinical-grade performance.
- Worked with **messy real-world clinical datasets** - data cleaning, quality checks, and rigorous train/validation splits to avoid leakage.
- Collaborated with a cross-functional team of engineers and clinicians under product-development standards.

Lecturer, *Dept. of Computer Science and Engineering, Bangladesh University of Business and Technology (BUBT)* 01/2023 – 08/2023 | Dhaka, Bangladesh

- Designed and taught undergraduate courses in machine learning and computer vision.

Machine Learning Engineer (Intern), *Pioneer Alpha Ltd.* 01/2021 – 08/2021 | Dhaka, Bangladesh

- Researched and implemented production ML algorithms; selected datasets and representations for real-world applications.

SELECTED PROJECTS

Adversarially Robust Few-Shot Anomaly Detection with Vision Foundation Models, *Accepted to BMVC 2026* 2026

- Differentiable distance surrogate exposing non-parametric k-NN detectors to gradient-based attack; two-level defense (PatchShift, FeaturePurifier) trained from the normal support set alone, leaving the backbone frozen. Matches full-shot adversarially trained baselines on pixel AUROC without adversarial training of the encoder.

Trust-Vista: Trustworthy Attention with Learnable Uncertainty for Vision Transformers Under Review
Submitted to IEEE Transactions on Image Processing (TIP)

- Closed-form moment propagation through Bayesian convolution, attention, normalization, and residual blocks, trained by variational inference under Gaussian and reparameterized categorical likelihoods. Evaluated on six benchmarks up to ImageNet-1k under Gaussian noise, FGSM, PGD, and C&W∞, with ECE / NLL / Brier calibration and inference-cost comparison against MC dropout, SNGP, and deep ensembles.

Measurement-Grounded Benchmark for Phase-Mask Image Restoration, *ONR Project* 2026

- Dual-camera acquisition pipeline (Basler / pypylon, software-triggered, raw 12-bit Bayer, beam-splitter-aligned pairs) producing a 3,600-image four-condition dataset across standard and phase-mask optics with laser on/off, plus a measured-PSF library spanning the mask's defocus range.

PUBLICATIONS

Adversarially Robust Few-Shot Anomaly Detection with Vision Foundation Models 2026
British Machine Vision Conference (BMVC) 2026
Akib M. Khan, Bartosz Krawczyk

AttResDU-Net: Medical Image Segmentation Using Attention-based Residual Double U-Net, *IEEE* 2023
Akib M. Khan, Alif Ashrafee, Fahim S. Khan, Md. Bakhtiar Hasan, Md. Hasanul Kabir
10.1109/IJCNN54540.2023.10191528 (IJCNN 2023)

Real-time Bangla License Plate Recognition System for Low Resource Video-based Applications, *IEEE* 2022
Alif Ashrafee, **Akib M. Khan**, Mohammad Sabik Irbaz
10.1109/WACVW54805.2022.00054 (WACVW 2022)

Rethinking Cooking State Recognition with Vision Transformers, *IEEE* 2022
Akib M. Khan, Alif Ashrafee, Reeshoon Sayera, Shahriar Ivan, Sabbir Ahmed
10.1109/ICCIT57492.2022.10055869 (ICCIT 2022)

TECHNICAL SKILLS

Robustness & Trustworthy ML
FGSM, PGD, C&W, AutoAttack; adaptive attacks (EOT, BPDA); gradient-masking diagnostics; adversarial training and purification; test-time adaptation and test-time prompt tuning; calibration and uncertainty (Platt scaling, ECE, MC-dropout, deep ensembles, SNGP, variational inference).

Modelling
Vision foundation models (CLIP, DINOv2, ViT); restoration transformers (Restormer, NAFNet); CNNs and U-Net variants; few-shot, semi-supervised, and continual learning; anomaly detection and memory-bank retrieval.

ML Engineering
PyTorch, TensorFlow, Keras, scikit-learn, XGBoost; multi-GPU training on 8×A100 clusters; CUDA environment and dependency debugging; FAISS; ablation studies, hyperparameter sweeps, and error analysis; reproducible evaluation harnesses.

Computational Imaging
PSF measurement and HDR merging; shift-variant convolution; radiometric and noise characterization (dark, glare, fixed-pattern); Zemax optical simulation; Basler / pypylon camera control and synchronized acquisition.

Data & Tooling
NumPy, Pandas, OpenCV, Matplotlib; large-scale image processing pipelines; Linux HPC workflows; Git; LaTeX; MySQL.

Programming an Languages
Python (primary), C/C++, MATLAB, shell. English (fluent), Bengali (fluent).